



CSE422 : Artificial Intelligence

Project Report

Project Title : Diabetes Prediction

Group : 4

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Introduction

The project is aimed at being able to predict whether an individual has a high probability of diabetes regarding basic health indicators. Diabetes is a long-term illness with severe complications in case of failure to detect it at early age. Through supervised machine learning, we educated various classification models on a vast real-life dataset and evaluated the performance. We also used an unsupervised algorithm (K-Means) to search the natural clustering of data.

Dataset Description

The information in the dataset includes demographics and clinical factors including gender, age, body mass index, hypertension, heart disease, smoking, HbA1c, and blood glucose. The target column is the diabetic (0 non-diabetic 1 diabetic).

This is a classification problem as the variable of interest diabetes takes two discrete cardinal values (0 and 1) or absence or presence of diabetes, instead of numeric values.

The overall size of the raw dataset is 100,000 rows and the number of features (not including the label) is 8. It is a binary classification problem and has a mixture of numeric and categorical attributes in it.

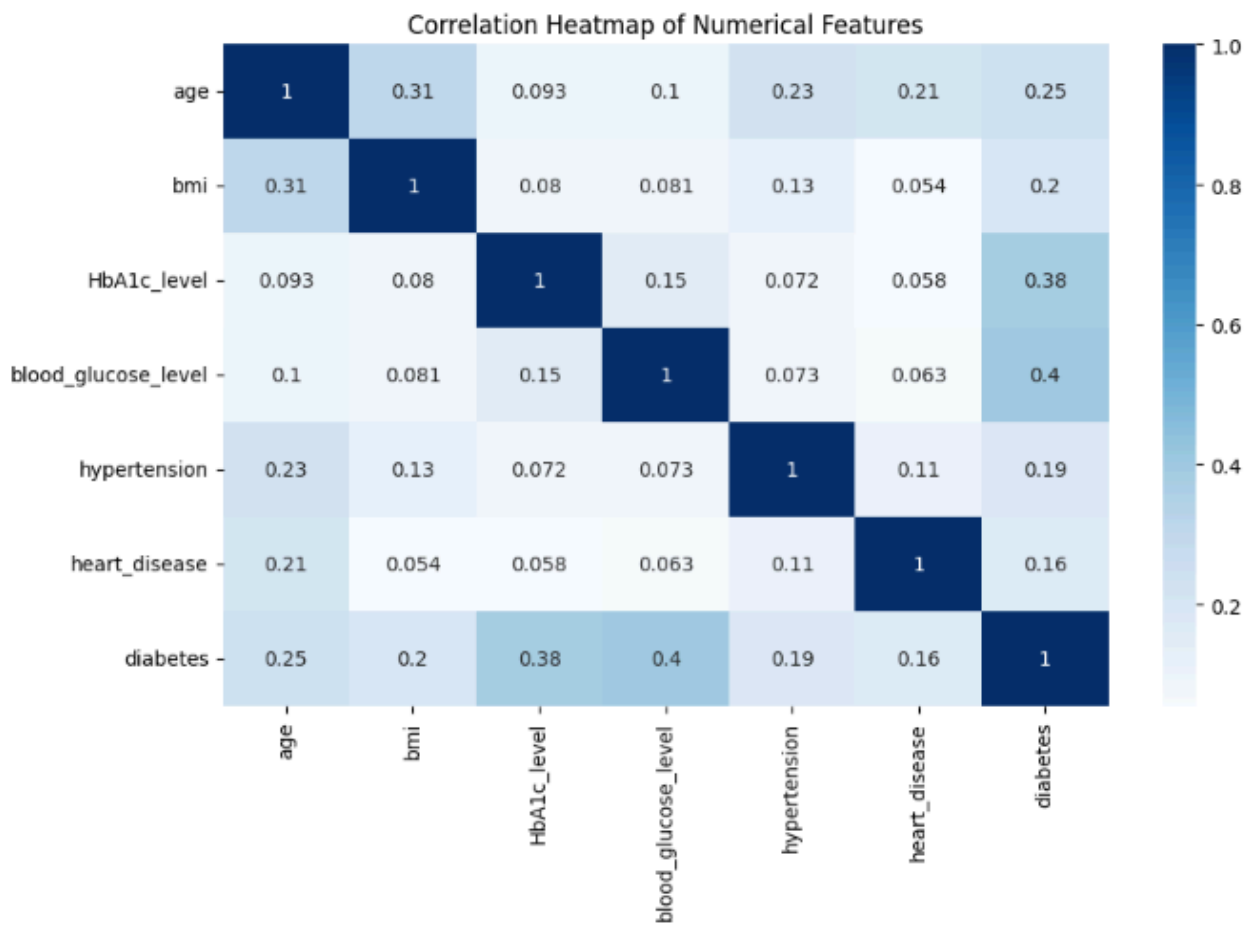
The dataset includes quantitative (numerical) data in terms of age, BMI, glucose level and categorical data in terms of gender and smoking history.

Yes, text categorical values should be encoded since machine learning engines are not able to take in text categorical values. Encoding translates

categories to the form of numbers to allow algorithms to extract patterns out of them.

Correlation of Numeric Features

Figure shows the correlation heatmap among numeric variables.



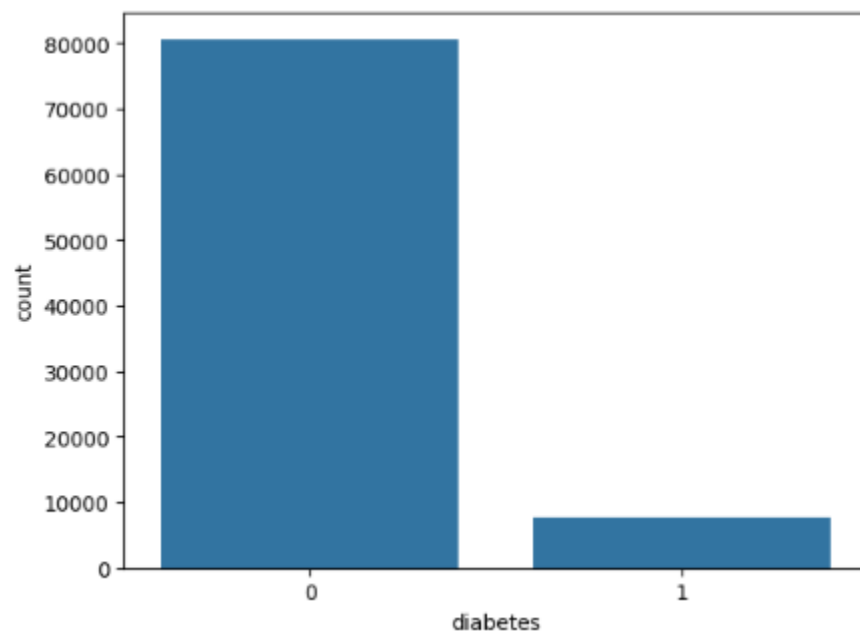
In the heat map, the blood glucose level and the HbA1c level demonstrate the positive correlation with diabetes, whereas the age and BMI features demonstrate the moderate correlation. It means that the features related to glucose have the most significant impact.

Class Imbalance

No, the dataset is imbalanced. The cases of the non-diabetic (class 0) exceed 80,000 which is overwhelmingly more than diabetic cases (class 1) amount to less than 10,000 meaning the proportion of classes in the output variable is unbalanced.

The imbalance was graphically represented as bar chart (countplot) of the target variable diabetes in which two bars symbolize the two classes (0 and 1) which clearly shows the prevalence of the non-diabetic class.

Figure illustrates the break even.

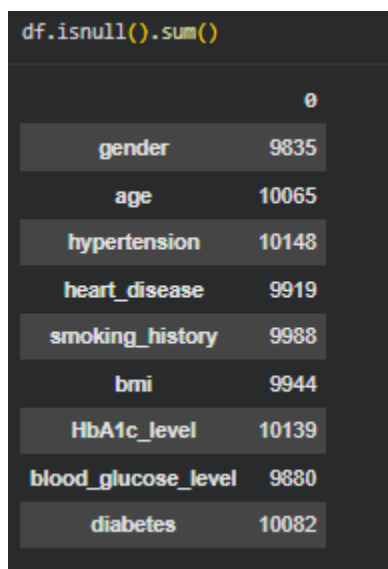


The results of EDA showed that there exists the strong positive relationship between diabetes and the level of blood glucose and HbA1c. The sample is unbalanced since the proportion of non-diabetic cases prevails. The other characteristics like age and BMI have moderate effects.

Dataset pre-processing

Null / Missing values

The data has both categorical and numbers with missing (null) values. The missing values cannot be trained in machine learning models and can result into errors or biased results.



```
df.isnull().sum()
```

	0
gender	9835
age	10065
hypertension	10148
heart_disease	9919
smoking_history	9988
bmi	9944
HbA1c_level	10139
blood_glucose_level	9880
diabetes	10082

From the figure we can see that there are many null values in every feature/column

Solution

- Duplicate rows were removed.
- To minimize the influence of the outliers, numerical characteristics were filled in with median values.
- Mode values were the ones used to provide categorical values.
- Rows that had no target value were deleted since the output labels are compulsory.

```

columnsNumerical = ['age', 'bmi', 'HbA1c_level', 'blood_glucose_level', 'hypertension', 'heart_disease']
for col in columnsNumerical:
    df[col].fillna(df[col].median(), inplace=True)

columnsCatagorical = ['gender', 'smoking_history']
for col in columnsCatagorical:
    df[col].fillna(df[col].mode()[0], inplace=True)

df.dropna(subset=['diabetes'], inplace=True)
df['diabetes'] = df['diabetes'].astype(int)

```

```
df.isnull().sum()
```

```

0
gender      0
age         0
hypertension 0
heart_disease 0
smoking_history 0
bmi         0
HbA1c_level 0
blood_glucose_level 0
diabetes     0

```

```
dtype: int64
```

Categorical values

Certain factors including gender and smoking history are categorical (written). The majority of machine learning algorithms have numerical input and thus cannot accept these features directly.

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 88183 entries, 0 to 99999
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   gender                 88183 non-null  object
1   age                   88183 non-null  float64
2   hypertension           88183 non-null  float64
3   heart_disease          88183 non-null  float64
4   smoking_history        88183 non-null  object
5   bmi                   88183 non-null  float64
6   HbA1c_level            88183 non-null  float64
7   blood_glucose_level    88183 non-null  float64
8   diabetes               88183 non-null  int64
dtypes: float64(6), int64(1), object(2)
memory usage: 6.7+ MB

```

Solution

Label encoding was used to convert categorical variables into numerals so that they could be correctly processed by the models without dimensionality increment.

```
le = LabelEncoder()

df['gender'] = le.fit_transform(df['gender'])
df['smoking_history'] = le.fit_transform(df['smoking_history'])
```

```
df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 88183 entries, 0 to 99999
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   gender                88183 non-null  int64
1   age                  88183 non-null  float64
2   hypertension          88183 non-null  float64
3   heart_disease         88183 non-null  float64
4   smoking_history       88183 non-null  int64
5   bmi                  88183 non-null  float64
6   HbA1c_level           88183 non-null  float64
7   blood_glucose_level   88183 non-null  float64
8   diabetes              88183 non-null  int64
dtypes: float64(6), int64(3)
```

Feature Scaling

The numerical characteristics are variable (e.g. age vs blood glucose level). Such models as the Logistic Regression and Neural Networks are feature-scale sensitive and this could influence the convergence and performance.

Solution

Standardization (StandardScaler) was used to normalize the numerical values in such a way that they have a mean of 0 and a variance of 1 without giving more weight to any features. Furthermore, feature scaling was performed following the division of the dataset in order to avoid any data leakage with the scaler only being fitted to the training dataset, and applied to the test dataset.

Dataset splitting

The dataset was split into a stratified random manner. Stratification gives an assurance that there is no difference in the level of classes in the target variable (diabetes) between training and test sets and this is relevant because of class imbalance.

The dataset was split into 70% training data and 30% testing data. This was done through model learning in the training set and testing the performance in the testing set without any form of bias.

```
x = df.drop('diabetes', axis=1)
y = df['diabetes']

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_state=42, stratify=y)
```

Model training & testing (Supervised)

We apply two important modes of supervised learning (training and testing the dataset) KNN, Decision Tree. These models have been chosen since they are appropriate to binary classification issues.

KNN (K-Nearest Neighbors)

KNN is an algorithm that is very simple and intuitive and is depended on the proximity concept. It is a lazy-learning and non-parametric algorithm, and implies no particular data distribution and no explicit model is constructed in training.

Rather, it memorizes the training data and decides a stopped data point in accordance with the higher classification among the nearest neighbors.

Decision Tree

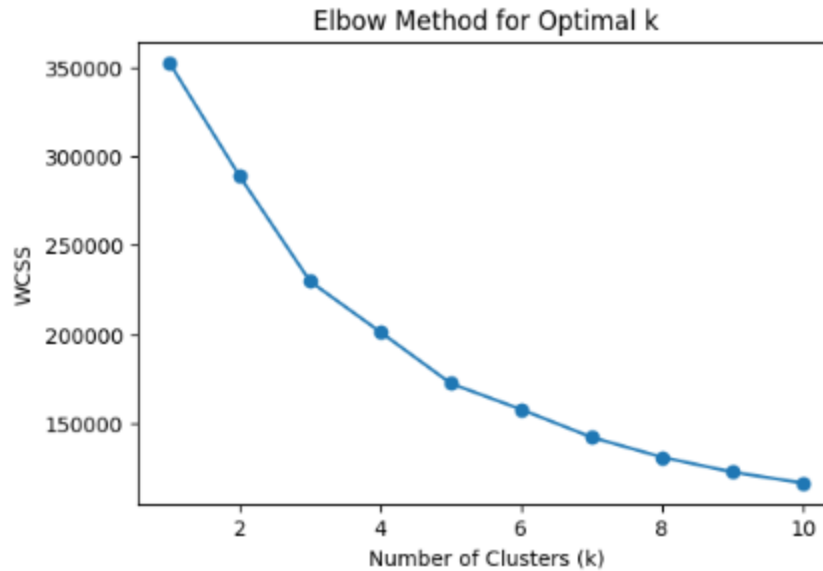
Decision Tree is a supervised learning model, which operates by recursively dividing the data according to the features. It is able to capture non-linear relations and gives tree-like structure the interpretation of which is easier because decisions are easier to see how they are made.

Neural Network

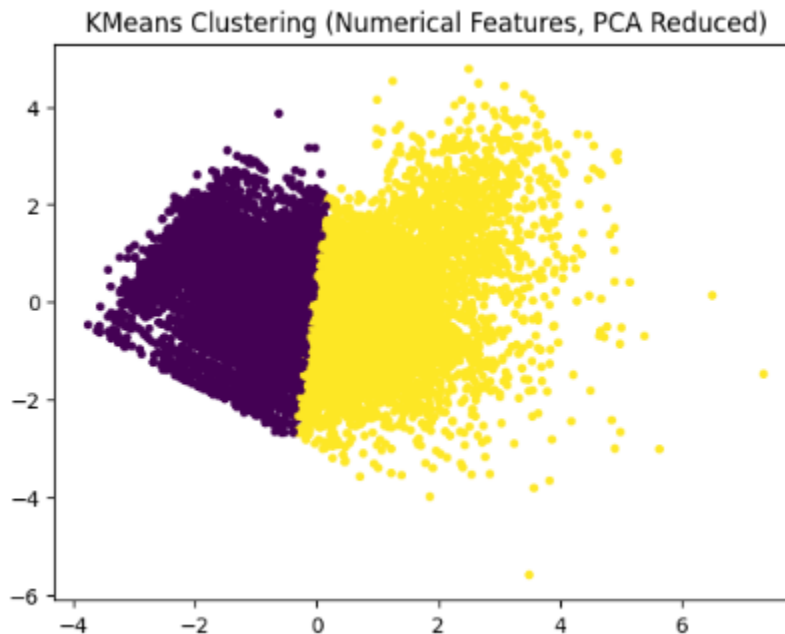
Another model employed on the data was a Neural Network because it can learn non-linear and complex relationships between the features. A Multi-layer Perceptron (MLP) was used in the model; this model is made up of more than one hidden layer and activation functions. Neural Networks work quite well with large data sets and high dimensional data. The Neural Network in this project was trained using a training set and tested using a test set in analyzing its classification performance.

K-Means Clustering (Unsupervised)

Besides the supervised model of learning, the problem was approached as a non-supervised learning problem to the K-Means clustering algorithm. K-Means is a distance based clustering where data are grouped into predetermined number of clusters by minimizing the within cluster distance sum squared of the the cluster centroid. Standardizing of numerical features was carried out prior to clustering and Elbow Method used to estimate an adequate number of clusters.



Following the location of the elbow point and the binary dataset, $k = 2$ was taken. To make the data two-dimensional, to visualize it, Principal Component Analysis (PCA) was applied.



The outcome of the clustering process indicates that a number of two major clusters were created based on the numerical aspects. The clusters are partly

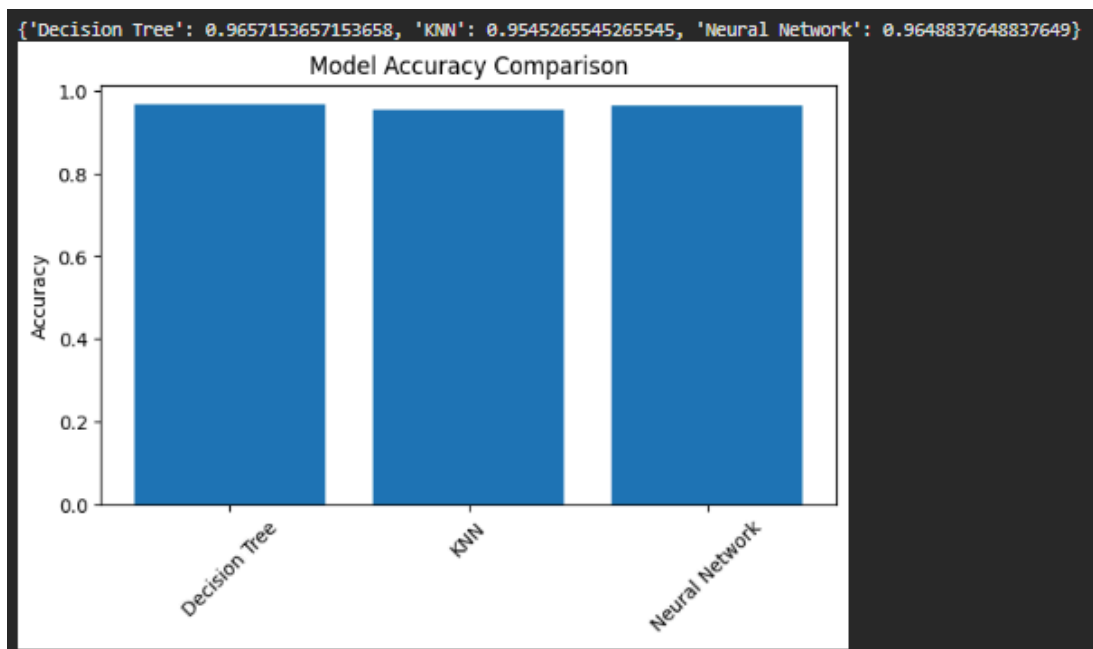
divided after reduction by PCA, which suggests that there are natural groupings in the information. That is, however, to some extent we see that classes of diabetes are not completely independent without supervision.

Model selection/Comparison analysis

In order to compare and assess the performance of the supervised classification models, the several evaluation metrics were employed such as accuracy, precision, recall, confusion matrix and ROC-AUC score. The metrics give a general overview of the model performance.

Accuracy Comparison

The comparison of prediction accuracy of all trained models was done using a bar chart. Decision Tree had the best accuracy then there was Neural Network and then there were KNN with relatively lower accuracy.



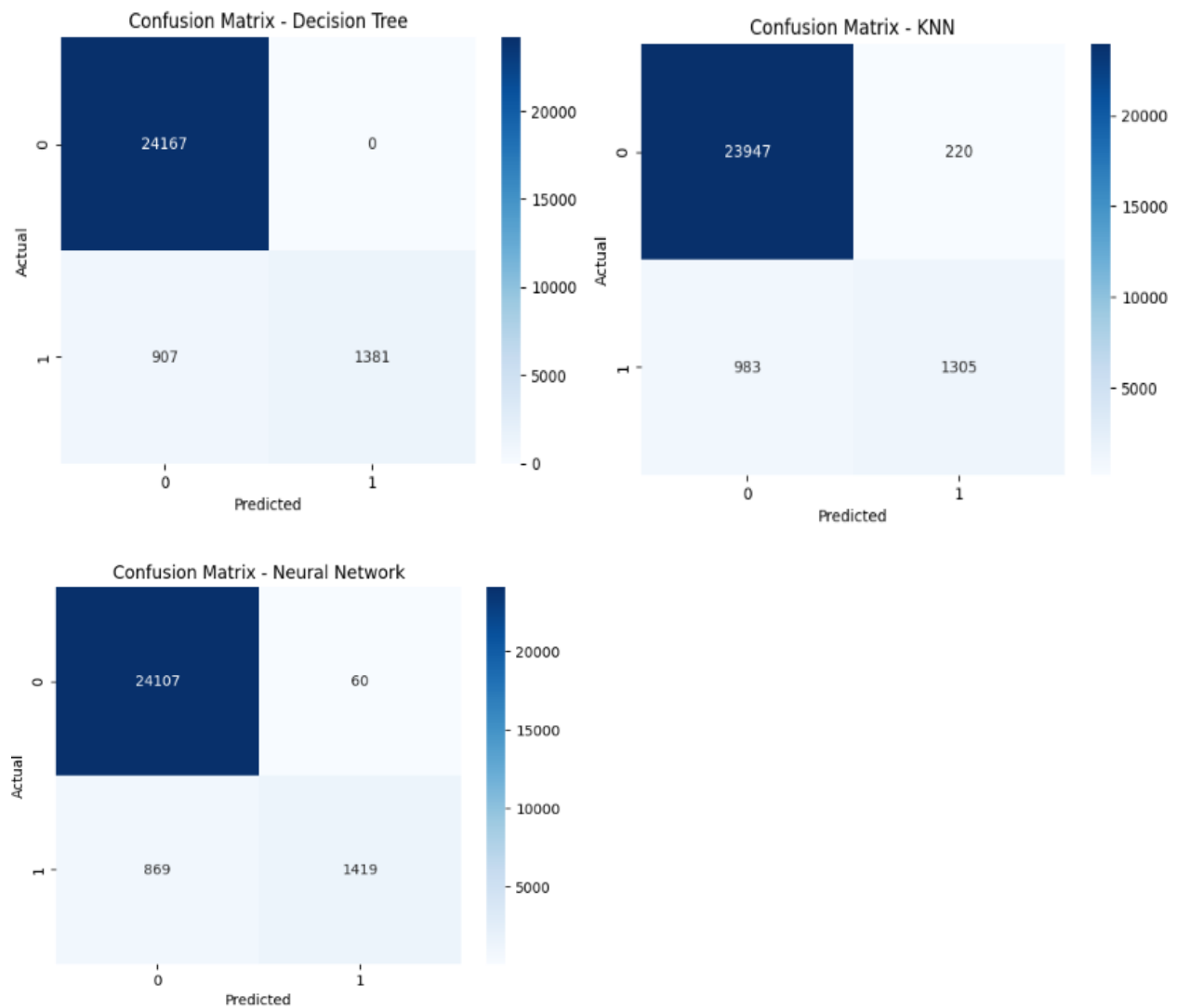
Precision and Recall Analysis

The highest precision of the Decision Tree was reached which means that the number of false positives was minimal, but the recall was lower because few instances of diabetic were missed. Neural Network displayed a higher accuracy to recall ratio, and it is more dependable. KNN obtained relatively low recall.

```
Decision Tree
Precision: 1.0
Recall: 0.603583916083916
F1-score: 0.7527936767511584
-----
KNN
Precision: 0.8557377049180328
Recall: 0.5703671328671329
F1-score: 0.6845003933910306
-----
Neural Network
Precision: 0.9713506139154161
Recall: 0.6223776223776224
F1-score: 0.7586574320724561
-----
```

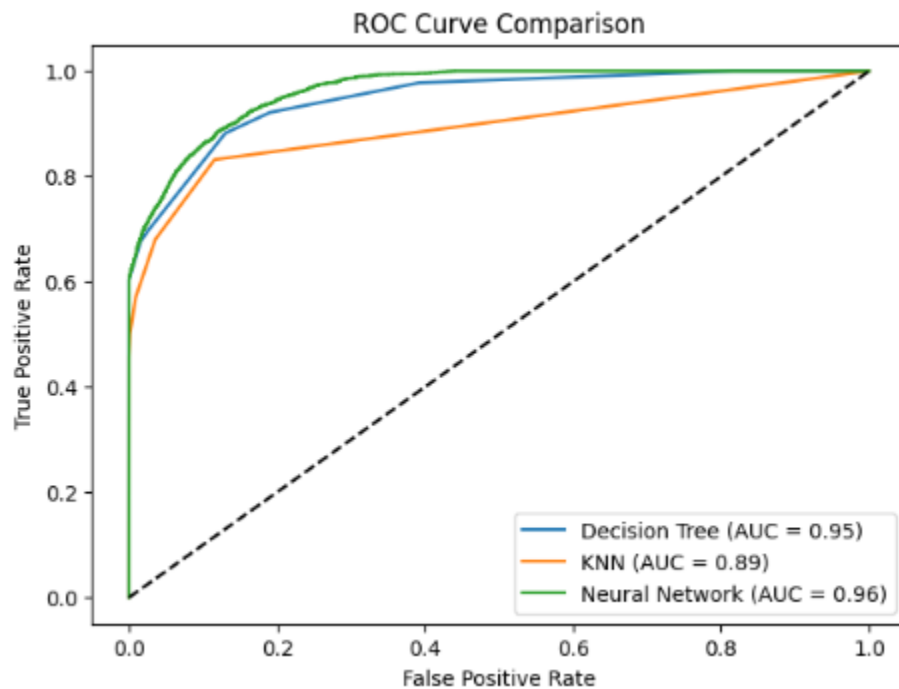
Confusion matrix

The case study of confusion matrices indicates that the Neural Network contained less false negatives, and thus, it is more dependable in the identification of diabetic cases. In general, the Neural Network was chosen to represent the best-performing model since it is more accurate and performs well in the classification.



ROC Curve and AUC Score

Each model was plotted on ROCs and AUC scores derived. The Neural Network had the highest AUC meaning that it is a better class separator than other models.



Conclusion

Due to the experimental outcomes, it is noted that the Neural Network model was the most effective amongst all the models that were tested, as it produced high accuracy as well as balanced precision and recall. This implies that the model is suitable in learning complicated and non-linear relationships in the dataset.

This good performance of the Neural Network can be explained with the (i) capability to operate with large data and (ii) ability to detect complex interactions of features whereas models like Decision Tree and KNN demonstrated weakness in overfitting and class imbalance. Proper preprocessing, scaling of features and stratified splitting were also used to enhance better model performance, assisted to reduce the bias and enhance the learning process. Simpler models, on the other hand, were more susceptible to the class imbalance and feature restriction.

The issues encountered during this project were the management of missing values, class imbalance, and choosing the right evaluation measurements, other than the accuracy. There was also a need to analyze the results of unsupervised clustering carefully and select the best number of clusters.

Comprehensively, the project shows that proper preprocessing, the selection of the adequate model, and an extensive evaluation is the key to building the credible machine learning models devoted to medical prediction.